Optimizing IT Support Team Performance Using Analytics

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Course:Virtual Internship 6.0-IT Support Team

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Project Overview

- This project focuses on analyzing IT support ticket data to identify performance trends, optimize resolution times, and improve service efficiency using data analytics techniques.
- The goal is to understand patterns in customer issues, ticket categories, and support performance metrics to enhance workflow and decision-making.

Objectives

- Analyze IT support ticket data

- Identify trends in ticket volume and priority

- Improve resolution efficiency

- Understand issue categories and clusters

- Provide insights for performance optimization

Dataset Description

- The dataset is obtained from an internal IT support ticket system. It contains details such as ticket type, priority, category, resolution time, and other related attributes.
- This data helps in analyzing the performance of IT support teams and identifying inefficiencies.

Tools Used

- Python

- Pandas

- NumPy

- Matplotlib / Seaborn

- Jupyter Notebook

Data Loading

The dataset was loaded using pandas library. Initial inspection was done to understand the structure of the dataset.

```
In [2]: import pandas as pd
        import numpy as np
        import matplotlib.pyplot as plt
```

```
In [3]: df = pd.read_excel("customer_support_tickets.xlsx")
```

```
In [4]: df.head()
```

Out[4]:

	Ticket ID	Customer Name	Customer Email	Customer Age	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	Ticket Subject	Unnamed: 9	Ticket Status	Resolution	Ticket Priority	Ticket Channel	First Response Time	Time to Resolution	Customer Satisfaction Rating
0	1	Marisa Obrien	carrollallison@example.com	32	Other	GoPro Hero	2021-03-22	Technical issue	Product setup	NaN	Pending Customer Response	NaN	Critical	Social media	2023-06-01 12:15:36	NaT	NaN
1	2	Jessica Rios	clarkeashley@example.com	42	Female	LG Smart TV	2021-05-22	Technical issue	Peripheral compatibility	NaN	Pending Customer Response	NaN	Critical	Chat	2023-06-01 16:45:38	NaT	NaN
2	3	Christopher Robbins	gonzalestracy@example.com	48	Other	Dell XPS	2020-07-14	Technical issue	Network problem	NaN	Closed	Case maybe show recently my computer follow.	Low	Social media	2023-06-01 11:14:38	2023-06-01 18:05:38	3.0
3	4	Christina Dillon	bradleyolson@example.org	27	Female	Microsoft Office	2020-11-13	Billing inquiry	Account access	NaN	Closed	Try capital clearly never color toward story.	Low	Social media	2023-06-01 07:29:40	2023-06-01 01:57:40	3.0
4	5	Alexander Carroll	bradleymark@example.com	67	Female	Autodesk AutoCAD	2020-02-04	Billing inquiry	Data loss	NaN	Closed	West decision evidence bit.	Low	Email	2023-06-01 00:12:42	2023-06-01 19:53:42	1.0

Data Understanding

- The dataset was explored to understand column names, data types, and relationships between variables.
- Functions like info() and describe() were used to analyze the dataset.

```
In [23]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8469 entries, 0 to 8468
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Ticket ID             8469 non-null   int64
 1   Customer Name         8469 non-null   object
 2   Customer Age          8469 non-null   int64
 3   Customer Gender       8469 non-null   object
 4   Product Purchased     8469 non-null   object
 5   Date of Purchase      8469 non-null   datetime64[ns]
 6   Ticket Type           8469 non-null   object
 7   Ticket Subject        8469 non-null   object
 8   Ticket Status         8469 non-null   object
 9   Ticket Priority       8469 non-null   object
 10  Ticket Channel        8469 non-null   object
 11  First Response Time   8469 non-null   int64
 12  Time to Resolution    8469 non-null   int64
dtypes: datetime64[ns](1), int64(4), object(8)
memory usage: 860.3+ KB
```

```
In [24]: df.describe()
```

Out[24]:

	Ticket ID	Customer Age	Date of Purchase	First Response Time	Time to Resolution
count	8469.000000	8469.000000	8469	8.469000e+03	8.469000e+03
mean	4235.000000	44.026804	2020-12-30 01:35:13.071201024	-1.945559e+18	-5.656600e+18
min	1.000000	18.000000	2020-01-01 00:00:00	-9.223372e+18	-9.223372e+18
25%	2118.000000	31.000000	2020-07-02 00:00:00	-9.223372e+18	-9.223372e+18
50%	4235.000000	44.000000	2020-12-31 00:00:00	1.685597e+18	-9.223372e+18
75%	6352.000000	57.000000	2021-07-01 00:00:00	1.685630e+18	1.685596e+18
max	8469.000000	70.000000	2021-12-30 00:00:00	1.685667e+18	1.685667e+18
std	2444.934048	15.296112	NaN	5.141027e+18	5.117725e+18

Missing Values Check

The dataset was checked for missing or null values to ensure data quality before analysis.

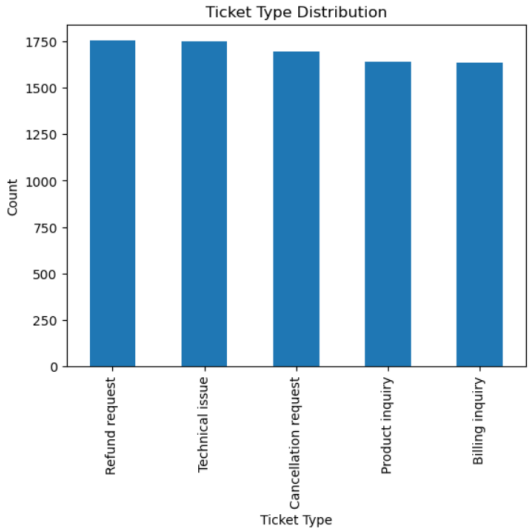
```
In [7]: df.isnull().sum()

Out[7]: Ticket ID                        0
        Customer Name                    0
        Customer Email                   0
        Customer Age                     0
        Customer Gender                  0
        Product Purchased                0
        Date of Purchase                 0
        Ticket Type                      0
        Ticket Subject                   0
        Unnamed: 9                    8469
        Ticket Status                    0
        Resolution                    5700
        Ticket Priority                  0
        Ticket Channel                   0
        First Response Time           2819
        Time to Resolution            5700
        Customer Satisfaction Rating  5700
        dtype: int64
```

Initial Analyzing

- Initial analysis was performed to understand ticket distribution based on type, priority, and category.
- This helps in identifying which types of issues occur most frequently.

```
In [38]: df['Ticket Type'].value_counts().plot(kind='bar')
         plt.title("Ticket Type Distribution")
         plt.xlabel("Ticket Type")
         plt.ylabel("Count")
         plt.show()
```



Data Cleaning

- Basic data cleaning steps were performed to handle missing or inconsistent values in the dataset.
- This ensures the dataset is ready for further analysis in upcoming milestones.

Conclusion

- In this milestone, the dataset was successfully loaded, explored, and prepared for analysis.
- Initial insights were obtained regarding data structure and ticket distribution.
- Further analysis and visualization will be carried out in the next milestones.